

Assignment SU3: Implementing a Text-Based Chess Survival Game with the MyArrayList Class

Objective:

Create a console-based Chess Survival game in JAVA where the player controls a King on an 8×8 chessboard, dodging advancing enemy pawns and collecting royal treasures. You will use the provided MyArrayList class to manage the enemy pawns and treasure chests on the board.

Instructions:

1. **Create an 8×8 game grid** (standard chessboard dimensions).
2. **Initialize the player:** Start the King K at the bottom-center of the grid (row 7, column 3).
3. **Manage enemy pawns with MyArrayList:** Use a MyArrayList<Point> to store pawn positions. At the end of each turn, spawn a new enemy pawn at row 0 in a random column. A pawn should not spawn on the King or any treasure.
4. **Manage treasures with MyArrayList:** Use a MyArrayList<Point> to store treasure positions. Place two treasures randomly on the grid initially. Treasures should not spawn on the King or any pawn. When the King collects a treasure, increment the score and spawn a new treasure at a random empty location.
5. **Allow the user to control the King:** Accept keyboard input (U = up, L = left, D = down, R = right). The King moves exactly one square in the chosen cardinal direction per turn, just like a real chess king.
6. **Process each turn:**
 - Move the King based on user input (clamp vertical movement between rows 0–7; wrap horizontal movement between columns 0–7).
 - Check if the King collected a treasure. If so, remove the treasure, add 10 points, and spawn a new one.
 - Check if the King captured a pawn by moving onto it. If so, remove the pawn and add 5 bonus points.
 - Move every existing pawn down by one row (pawns advance toward the King).
 - Check for collisions between pawns and the King.
 - Remove pawns that march off the bottom edge (row >= 8).
 - Spawn a new pawn at the top (row 0, random column).
7. **Check for game over conditions:**
 - Any pawn occupies the same cell as the King after moving.

8. **Display the grid:**

- Show King as K.
- Show enemy pawns as P.
- Show treasures as *.
- Show empty cells as ..

9. **Add a scoring system:** Award 10 points per treasure collected and 5 points per pawn captured. Display the score each turn.

10. **End the game when the King is captured:** Display the final score.

11. **Allow the player to restart the game after game over:** Prompt Play again? (Y/N).

12. **Implement boundary wrapping for horizontal movement:** Moving left from column 0 wraps to column 7, and moving right from column 7 wraps to column 0. Vertical movement should be clamped (the King cannot move above row 0 or below row 7).

Compile and run ChessSurvivalTest.java to ensure the game works as expected. Test the program with different actions to ensure proper handling of edge cases.